

Relational Schema and SQL script

Employees(ssn : NUMERIC, first_name : VARCHAR, last_name : VARCHAR, email : VARCHAR, salary : DECIMAL)

```
CREATE TABLE Employees (  
    ssn NUMERIC(9,0),  
    first_name: VARCHAR(255),  
    last_name: VARCHAR(255),  
    email: VARCHAR(255),  
    salary: DECIMAL(10,2),  
    PRIMARY KEY(ssn)  
)
```

Baristas(ssn : NUMERIC, day : DATE, start_time : TIME, end_time : TIME) with foreign key (ssn) to Employees(ssn)

```
CREATE TABLE Baristas (  
    ssn NUMERIC(9,0),  
    day DATE,  
    start_time TIME,  
    end_tme: TIME,  
    PRIMARY KEY(ssn),  
    FOREIGN KEY(ssn) REFERENCES Employees(ssn)  
)
```

Managers(pertentage_ownership : NUMERIC) with foreign key (ssn) to Employees(ssn)

```
CREATE TABLE Managers(  
    ssn NUMERIC(9, 0),  
    percentage_ownership NUMERIC(5,2),  
    FOREIGN KEY(ssn) REFERENCES Employees(ssn)  
)
```

Manages(ssn: NUMERIC, name: TEXT) foreign key (ssn) to Managers and foreign key(name) to Inventory_management(name)

```
    ssn NUMERIC(9,0),  
    Name TEXT  
    FOREIGN KEY(ssn) REFERENCES Managers(ssn)  
    FOREIGN KEY(Name) REFERENCES Inventory_management(name)
```

Inventory_management(name: TEXT, unit : TEXT, quantity : INT, price : DECIMAL) with foreign key (ssn) to Managers(ssn)

```
CREATE TABLE Inventory_management(  
    name TEXT NOT NULL,  
    unit TEXT,  
    ssn NUMERIC(9,0),  
    quantity INTEGER CHECK (quantity >= 0),  
    price DECIMAL(10, 2) CHECK (price > 0.0),  
    PRIMARY KEY(name),  
    FOREIGN KEY(ssn) REFERENCES Managers(ssn),  
)
```

–used for payment_method in sales

```
CREATE TYPE payment_method as ENUM(cash, credit card, app)
```

Sales(time : TIME, day: DAY, quantity : DECIMAL, drink : VARCHAR, payment_method : payment_method) with foreign key (drink) to Menu(name) and foreign key (time, day) to Promotions(time, day)

```
CREATE TABLE Sales(  
    time TIME,  
    day DAY,  
    quantity DECIMAL (10),  
    drink VARCHAR (30),  
    payment_method payment_method,  
    PRIMARY KEY(time, day),  
    FOREIGN KEY(drink) REFERENCES Menu(name),  
    FOREIGN KEY(time, day) REFERENCES Promotions(time, day)  
)
```

Accounting(day : DATE, time : TIME, account_balance : DECIMAL) with foreign key (day, time) to Sales(day, time)

```
CREATE TABLE Accounting (  
    day DATE,  
    time TIME,  
    account_balance DECIMAL(20, 2),  
    PRIMARY_KEY (day, time),  
    FOREIGN KEY(day, time) REFERENCES Sales(day, time))  
)
```

Promotions(day: DATE, time: TIME, menu: VARCHAR, promotion_price: DECIMAL)(
 CREATE TABLE Promotions
 day DATE,
 time TIME,
 menu VARCHAR(30) NOT NULL,
 promotion_price DECIMAL(10,2),
 PRIMARY KEY(day, time, menu)
)

Menu(name : VARCHAR, size : DECIMAL, type : VARCHAR, price : DECIMAL, hot_cold :
 VARCHAR)
 CREATE TABLE Menu (
 name VARCHAR(30),
 size DECIMAL(5,2),
 type VARCHAR(20),
 price DECIMAL(10,2),
 hot_cold VARCHAR(4),
 CHECK (hot_cold IN ('hot', 'cold')),
 CHECK (type IN ('tea', 'coffee', 'softdrink'))
 PRIMARY KEY(name)
)

Recipes (recipe_name: TEXT, ingredient_name: VARCHAR, ingredient_quantity: DECIMAL,
 ingredient_unit: VARCHAR, position_number: TINYINT, execution_description: TEXT) with
 foreign key (ingredient_name) from inventory_management(name) and foreign key
 (recipe_name) from Menu(name)
 CREATE TABLE Recipes
 recipe_name TEXT,
 ingredient_name VARCHAR(20),
 ingredient_quantity DECIMAL(20,2) CHECK (ingredient_quantity > 0),
 ingredient_unit VARCHAR(5),
 position_number TINYINT,
 execution_description TEXT,
 PRIMARY KEY (recipe_name, position_number),
 FOREIGN KEY (ingredient_name) REFERENCES Inventory_management(name),
 FOREIGN KEY(recipe_name) REFERENCES Menu (name)
)